

Cancer Research Society

Breaking Down Two Groundbreaking Oncology Studies — How Scientists Asked the Right Questions & Found Answers

Welcome to this edition of our Cancer Research Society breakdown! Today we're exploring two cutting-edge studies published in 2026 — one asking *when* we give cancer treatment, and another asking *what* we combine. Both show how smart study design leads to real breakthroughs.

STUDY #1

Does Timing of Immunotherapy Actually Matter?

The Big Question

Researchers noticed that our immune system follows a daily rhythm — it's more active at certain times of day. So they asked: **if immune checkpoint inhibitors work by boosting the immune system, does giving them earlier in the day make them work better?**

Why This Is Innovative

Most cancer treatment schedules are designed around *convenience*, not biology. This study challenged that assumption by looking at the body's internal clock — called the **circadian rhythm** — and asking whether it could be a hidden variable in treatment success. It's a completely new lens for thinking about drug delivery.

How Did They Build the Study?

Published in *JAMA Network Open* by Inoue et al., this was a **systematic review and meta-analysis** — meaning the researchers didn't run new experiments. Instead, they collected and combined data from many existing studies.

01

Search the Literature

Researchers searched MEDLINE (PubMed), Embase, and Web of Science for studies comparing **early vs. late** immunotherapy administration.

02

Screen & Select Studies

29 studies with 6,129 patients met the criteria — including randomized trials and cohort studies across multiple cancer types.

03

Extract & Assess Data

Two independent reviewers extracted data and checked for bias. They used **random-effects models** to combine results fairly.

04

Measure Outcomes

Primary outcomes: **Overall survival** and **progression-free survival** — the two most important metrics in oncology trials.

KEY FINDINGS

What the Timing Study Found

The results were striking. Earlier administration of immune checkpoint inhibitors was consistently linked to better survival outcomes across multiple cancer types.

0.60

Overall Survival HR

A hazard ratio below 1.0 means fewer deaths — early timing cut the risk significantly (95% CI: 0.51–0.70).

0.62

Progression-Free HR

Early dosing also delayed cancer progression (95% CI: 0.54–0.71) across the pooled patient population.

29

Studies Analyzed

Including 1 randomized trial, 1 prospective cohort, and 27 retrospective cohort studies across 10+ cancer types.

6,129

Patients Included

Spanning NSCLC, melanoma, gastric, renal cell carcinoma, biliary tract cancer, and more.

Which Cancers Benefited Most?

Subset analyses showed that the timing advantage was especially pronounced in five cancer types. The hazard ratios below all favor **earlier** immunotherapy administration.

Non–Small Cell Lung Cancer (NSCLC)

Overall survival HR = **0.58** — the strongest evidence, including data from the only randomized clinical trial in the meta-analysis.

Small Cell Lung Cancer

Overall survival HR = **0.37** — the most dramatic benefit seen in any subgroup, suggesting timing may be especially critical here.

Gastric Cancer

Overall survival HR = **0.61** and progression-free HR = 0.62 — consistent benefit across both endpoints.

Renal Cell Carcinoma & Biliary Tract Cancer

Both showed meaningful improvements in overall and progression-free survival, reinforcing the broad applicability of timing as a factor.



Takeaway for Club Members: A "hazard ratio" (HR) below 1.0 means reduced risk of death or progression. The closer to 0, the bigger the benefit!

STUDY #2

Combining Two Weapons Against Prostate Cancer

The second study — the **EORTC 1333/PEACE-3 trial** — asked a completely different question: *Can we fight bone-dominant prostate cancer better by combining two existing drugs?* This was a massive, multinational Phase III clinical trial with over 58 months of follow-up.

THE SCIENCE BEHIND IT

What Are These Two Drugs & Why Combine Them?

Enzalutamide

An **oral androgen receptor inhibitor** — it blocks testosterone from fueling prostate cancer cells. It's already the most commonly used first-line treatment for metastatic castration-resistant prostate cancer (mCRPC). Approved and proven, but cancer eventually finds ways around it.

Radium-223

An **alpha-particle emitting radioactive agent** that mimics calcium — it travels directly to bone where it delivers targeted radiation to bone metastases. FDA-approved in 2013 specifically for bone-dominant mCRPC. The innovation here was combining it *earlier* with enzalutamide rather than saving it for late-stage disease.

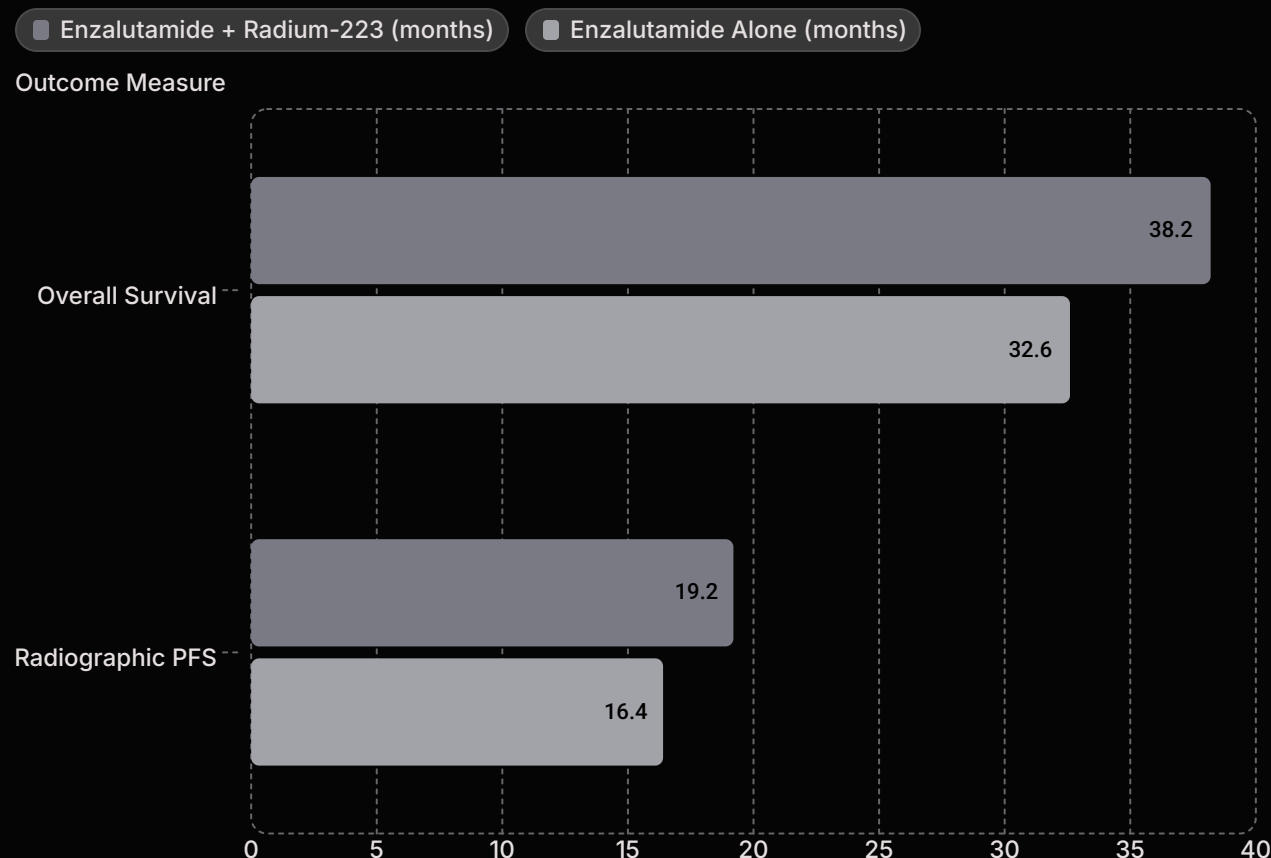


The Research Question: If enzalutamide targets the cancer cells and radium-223 targets the bone tumors directly, could using them together be significantly better than either one alone?

PEACE-3 RESULTS

The Survival Numbers That Mattered

After a median follow-up of **58 months** and 317 total deaths recorded, the EORTC 1333/PEACE-3 final analysis delivered compelling results across 446 patients with bone metastases.



The combination reduced the risk of prostate cancer mortality by **24%** (HR = 0.76, P = .0096) — a statistically significant and clinically meaningful result. At the 36-month mark, **54.2%** of combination patients were alive vs. **47.4%** in the single-drug arm.

What Are the Trade-Offs?

Every effective treatment comes with side effects. Understanding the risk-benefit balance is a core skill in reading research critically.

Side Effects to Know

- **Grade 3+ adverse events:** 69.3% (combo) vs. 57.6% (enzalutamide alone)
- **Hypertension:** ~36% in both groups — the most common toxicity
- **Osteonecrosis of the jaw:** 14 cases in combo arm vs. 3 in monotherapy arm (6.4%)
- **Rare blood cancers:** One case each of MDS, AML, and CML — all in the combo arm

Study Limitations

- Many patients in the study had *not* previously received an androgen receptor pathway inhibitor (ARPI) — less representative of today's patients
- Modern standard of care often includes ARPIs before mCRPC, so real-world applicability is still being evaluated
- Crossover to the combination arm was not permitted, which some critics noted
- Subgroup benefits were reduced in older patients (≥ 75 years) and those with prior docetaxel

What Can We Learn From Both Studies?

These two studies are incredible examples of how asking the *right question* — and designing a study carefully to answer it — can shift how doctors treat cancer patients worldwide.



Timing Is a Variable

The circadian rhythm study shows that *when* you give a drug can be just as important as *what* drug you give. Biology isn't one-size-fits-all.



Combination Strategies Work

PEACE-3 demonstrated that combining drugs with different mechanisms — one hormonal, one radioactive — can meaningfully extend lives by attacking cancer from two angles.



Study Design Matters

A meta-analysis pools data for bigger conclusions. A randomized trial like PEACE-3 controls for bias. Both methods are essential tools in the research toolbox.



More Research Needed

Both studies call for future prospective trials to confirm causation. Science is a process — findings open doors rather than close them.



✓ For Our Club: Next time you read a study, ask yourself — What was the question? How was it designed? Who were the patients? What are the limitations? That critical thinking *is* scientific thinking.